AMRIT GAUTAM

Embedded Systems & FPGA Engineer — B.S. Computer Engineering, UT Arlington (Dec 2026)
Dallas–Fort Worth, TX · mynameisamritgautam@gmail.com · 945-209-4242
linkedin.com/in/mynameisamritgautam · github.com/mynameisamritgautam · mynameisamritgautam.github.io

EDUCATION

University of Texas at Arlington — B.S. Computer Engineering Expected Dec 2026
- Relevant coursework: Digital Logic I–II, Embedded Systems I–II, Operating Systems, Signal Processing, Algorithms & Data Structures, Engineering Probability

Dallas College — Associate of Science Aug 2023
- 3.74 GPA · Phi Theta Kappa honor society

PROJECTS

5-Stage Pipelined RISC-V CPU Verilog · Xilinx Zynq-7000S · Vivado
- Designed and implemented a pipelined RISC-V processor (fetch/decode/execute/memory/writeback) with hazard detection, stalling, and data forwarding
- Debugged BRAM inference and a combinatorial loop at synthesis; verified on-chip behavior with an Integrated Logic Analyzer on real FPGA hardware

IGVC Autonomous Ground Vehicle — Senior Design NVIDIA Jetson · LiDAR · Team of 7
- Contributed to UTA's Intelligent Ground Vehicle Competition entry: a self-driving vehicle integrating Jetson compute, LiDAR, camera vision, and GPS
- Installed and wired the Sabertooth 2x32 motor drive system, rebuilt power/signal wiring for reliability, tuned lane-detection and autodrive behavior, and prototyped the NRF24 wireless emergency stop

Parallel Prefix Sum on GPU — CUDA CUDA C/C++ · Team of 2
- Implemented a work-efficient parallel scan (Blelloch), progressing from a sequential CPU baseline to a naive kernel to a shared-memory version with coalesced reads; course benchmarks measured ~40x over the sequential baseline
- Semester of CUDA lab work: multidimensional grids, shared-memory benchmarking, convolutions, reductions, parallel scan

Embedded Ethernet Stack with DHCP Client Embedded C · TM4C123 · ENC28J60
- Implemented a DHCP client state machine (DISCOVER→OFFER→REQUEST→ACK, lease renewal and rebinding) over a raw Ethernet controller via SPI — no OS, no vendor network stack
- Built a UART command-line interface for configuration and EEPROM persistence so settings survive power cycles

Closed-Loop DC Motor Control System STM32F446RE · Team of 3
- Owned component selection and complete hardware integration for a 1 kHz PID speed-control system: TB6612FNG H-bridge, quadrature-encoder motor, ACS712, SSD1306, HC-SR04, and SD logging — full pin mapping (PWM, SPI, I2C, ADC, encoder timer), 12 V power with shared ground, and breadboard bring-up; teammates built the PID controller and firmware

Automated Content Production Pipeline Python · REST APIs · OAuth 2.0
- Built an end-to-end Python system orchestrating five external services: script generation (Claude API), voiceover (ElevenLabs), footage (Pexels API), video assembly (MoviePy), and scheduled uploads (YouTube Data API, OAuth 2.0) — running in production at ~1 automated video per day

EXPERIENCE

Trinity Phone Repair — Summer Intern May 2022 – Aug 2022
- Diagnosed and repaired 50+ mobile devices across hardware and software faults; handled customer-facing intake on a 5-person team

SKILLS

- **Languages:** C, Python, CUDA C/C++, Verilog/SystemVerilog, C++, SQL, Bash, JavaScript/HTML/CSS
- **Embedded & FPGA:** ARM Cortex-M (STM32, TM4C123), Xilinx Zynq-7000 (Vivado: synthesis, timing, ILA), Intel MAX 10 (Quartus Prime), SPI, I2C, UART, IR, Ethernet (DHCP/UDP/IP/ARP), oscilloscope & logic-analyzer debugging
- **Software & Tools:** Git, Linux, Keil µVision, MATLAB, NumPy/pandas/matplotlib, REST APIs, OAuth 2.0